

Capstone Project - Planning Document

Prepared by: Abhinandan Mishra

Project Overview

Website	demo.nopcommerce.com	Framework	Playwright v1.4x+ JavaScript Node.js
Domain	E-Commerce B2C	Pattern	Page Object Model (POM) + Custom Fixtures
Test Cases	120+ (8 modules x 15 each)	Reporting	Allure Report + Playwright HTML Report
AI Tools	GitHub Copilot + ChatGPT	CI/CD	GitHub Actions / Jenkins (headless execution)

Why demo.nopcommerce.com?

Feature-Rich E-Commerce	Industry Standard UI	Complex Test Scenarios
Contains core e-commerce functionalities like Wishlists, Product Comparison, and multi-tier categories.	Mimics a highly realistic, production-grade online storefront with standard web elements.	Allows for testing complex states, such as guest checkout vs. registered user checkout.

Services & Test Case Breakdown (8 × 15 = 120 Test Cases)

#	Service / Module	Key Test Scenarios	Cases
1	Authentication	Valid login, invalid login, logout, registration with valid data, registration with existing email, password recovery, session persistence, empty fields validation, guest vs. registered states, password complexity rules, locked account simulation, redirection after login, UI validation of auth pages.	15
2	Product Catalog	Search by exact name, partial search, search with no results, filter by category (e.g., Computers, Apparel), sort by price (Low to High / High to Low), grid vs. list view toggle, pagination,	15

#	Service / Module	Key Test Scenarios	Cases
		breadcrumb navigation, product detail page UI, out-of-stock validation, price accuracy.	
3	Shopping Cart	Add single item, add multiple items, update quantity in cart, remove items, empty cart state, apply discount code, apply gift card, estimate shipping popup, cart badge update, cart persistence after login, subtotal/total calculation accuracy, tax display, continue shopping button.	15
4	Wishlist & Compare	Add to wishlist from catalog, add to wishlist from product page, move item from wishlist to cart, remove from wishlist, share wishlist link, add to compare list, compare attributes table validation, clear compare list, duplicate item handling in wishlist, empty states for both features.	15
5	User Profile	Update first/last name, update email, change password, view order history, add new billing address, edit existing address, delete address, required field checks, downloadable products tab validation, reward points display, back-in-stock subscriptions, newsletter toggle.	15
6	Checkout Process	Guest checkout flow, registered checkout flow, billing address selection, shipping method selection (Ground, Next Day), payment method selection (Credit Card, Check), order confirmation screen, order ID generation, tax recalculation at checkout, invalid card handling, empty cart redirect.	15
7	Customer Support & UI	Contact Us form submission, required field validation on forms, invalid email format handling, currency change toggle (USD/Euro), language selection, footer link validation, newsletter subscription form, recently viewed products section, responsive UI checks (mobile viewport).	15
8	E2E & Network Mocking	Full End-to-End purchase flow, visual regression on homepage, API mocking for specific categories (e.g., forcing a 500 error on product load), intercepting cart requests, modifying response bodies for out-of-stock items, testing network timeout handling, trace viewer generation on failure.	15

Implementation Milestones

Milestone	Deliverable
M1 - Setup & Config	Node.js workspace initialized, playwright.config.js with cross-browser profiles (Chromium, Firefox, WebKit), package.json scripts (test:headed, test:ui, test:report).
M2 - Base POM & Locators	Core Page Object classes created (LoginPage, HomePage, CartPage). Implementation of robust locators (getByRole, getByText), and data-driven testing via JSON files for user credentials and product data.
M3 - Advanced UI & Actions	Handling dynamic dropdowns, hovering over mega-menus (Categories), manipulating the shopping cart table, handling browser dialogs (alerts on deleting items), and multi-tab/window management.
M4 - E2E & Assertions	Linking Page Objects to create complete End-to-End workflows (e.g., Login → Search → Add to Cart → Checkout). Implementation of expect.soft() for non-critical UI checks.
M5 - Reporting & CI/CD	Integration of Allure Reports for trend analysis. Configuration of Playwright HTML reports. Setup of failure artifacts (screenshots, videos, trace files). Basic CI/CD pipeline setup (e.g., GitHub Actions workflow file).
M6 - Mocking & Polish	Using page.route() to intercept and mock backend responses to test edge cases without relying on the live demo server. Final code refactoring, adding JSDoc comments, and README documentation.